ZOO-102 Current papers fall 2020-2021

Q1: Reproductive Patterns

• The pattern of egg laying and development of young ones. • In all invertebrates and vertebrates one of the three patterns.

• Oviparity

• Ovoviviparity

• Viviparity

Q2: Define oviparous animals? 2marks

Oviparous

Which lay eggs and development outside the body of animals • Fertilization may be internal or external • Majority of invertebrates and many vertebrates are oviparous • Parental care may be or may not

Examples: Frogs, snakes, lizards, hens, duck, fishes, shark, penguins, butterflies, octopus,etc.

Q3: Define ovoviviparous animals? 2marks

Ovoviparous

Which eggs are retained into the body but the young one gets all its food from yolk of egg • Fertilization is internal • Development also internal

Examples: Invertebrates: Annelids, brachiopods, gastropods • Vertebrates: Fishes and reptiles, insects, rays, sharks, etc.

Q4: Define viviparous animals? 2marks

Viviparous

Which eggs develops into the body and the young one gets its nourishment from mother . Fertilization is internal. Development also internal

Examples: Invertebrates: Scorpions • Vertebrates: most mammals and elasmobranch fishes, few amphibians and reptiles, cattle, some shark, giraffes, bears,etc.

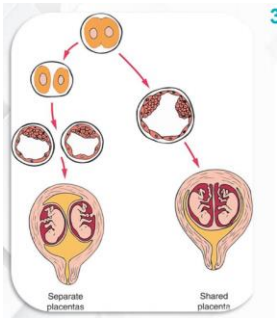
Q5: Difference between holometabolous Metamorphosis AND Hemimetabolous Metamorphosis)

Holometabolous Metamorphosis	Hemimetabolous Metamorphosis

This is also called complete metamorphosis • In about 88% insects • It is the type of metamorphosis which separates the physiological process of growth from differentiation and sexual stages Each process is completed without intervention of other stage	• It is also called gradual incomplete metamorphosis
Four stages • Egg • Larva • Pupa • Adult **• E.g., Insects**	• It is the mode of development of certain insects with three distinct stages • Egg • Nymph • Adult **• E.g., Aphid, grasshopper, cockroach**
Egg Stage • First stage of insect life cycle • Development starts after fertilization or sometime without fertilization • Early embryonic development completed at this stage • Egg stage is very short but in some cases undergo diapause	**Egg Stage** • First stage of development • Initial stages of development are completed at egg stage • Egg stage varies in different groups but in some cases undergo diapause • Eggs of aphids are hatched before they are layed
Larval Stage • Second stage of holometabolous life cycle • Larvae are hatched out from eggs • Motile and worm like form • Actively feed and grow in size • Never possess wings • Simple eyes • There are several larval stages or instars • Larva molts for passing from one instar to the next one	**Nymph Stage** • Nymph is an immature form hatched out from the eggs • Nymph lacks wings • They have underdeveloped reproductive organs • Nymph undergo gradual changes called instars • Aquatic insect may have tracheal gills
Pupa Stage • Third stage of holometabolous metamorphosis • Larva undergo metamorphosis to become pupa • Larva form a case or cocoon around itself • It is quiescent, non feeding developmental stage • Mostly pupa move very little, however some are mobile • Some insects undergo diapause as pupa • It is the stage of drastic changes in morphology and physiology	
Adult Stage • Final stage of metamorphosis • Adults have wings and reproductive organs • Most insects adult grow very little • Some do not feed and entirely focus on mating and reproduction	**Adult Stage** • The apparent difference from nymph is presence of wings • They are also reproductively mature

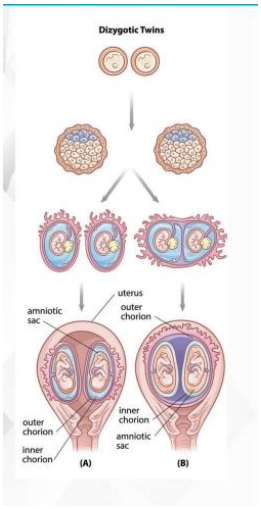
Q6: Define Monozygotic Twins

Produced from one zygote (identical twins). They are strikingly alike and of same sex. About 33% have separate placenta. Majority have separate amniotic sac. Small proportion have one placenta and one amniotic sac. Indicate separation occur after 9 day of pregnancy. Conjoining risk, condition is Siamese twining



Q7: Dizygotic Twins.3

Produced from two zygote (Fraternal twins). Do not have striking similarities. May be of same or different sex. They have separate placenta and separate amniotic sac



Q8: Three Functions Of Epididymis Of Human Male Reproductive organ.3

• Sperms from seminiferous tubules comes into a network of called rete testis

• Then into epididymis where:

• Sperms are store

• Muscular contraction propel sperm forward

Q9: Difference b/w synchrony and Asynchrony?

• Initially all the cells of embryo divide at the same time this is called synchrony

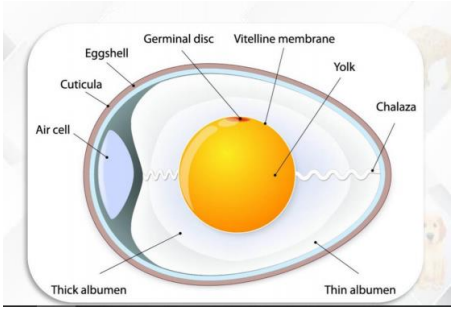
• As development proceeds, all cells don’t divide simultaneously. This is called asynchrony

Q10Define human population growth 5 marks

1. Population Growth • It is the increase in number of individuals in a population • Expanding population is the root cause of most environmental problems. • More population means more need for resource: food, urbanization, industrialization
2. 2. Earth's Carrying Capacity • How many human can be supported by earth? • Virtually all scientists agree that even 5 billion is more for earth to support with all facilities • Maximum carrying capacity of earth is may be 9 to 10 billion people
3. 3. Age Structure • Age structure shows the proportion of individuals at prereproductive, reproductive and postreproductive stage • Developing countries have more infants means high birthrates • In contrast to human, in natural population high birthrate is accompanied with high mortality

Q11: What is avigan egg 3 or 5 marks

Structure of Avian Egg

1. **Egg Formation**
   • In ovaries • Large amount of yolk • Mature egg released from ovaries • Comes into oviduct • Fertilization takes place in upper portion of oviduct
2. **Egg White**
   • Membranes and fluid collected around the egg while moving through oviduct • Water and albumin is deposited around the egg • This protect from desiccation and mechanical damage • Albumin is used for protection and source of nutrition • Two denser strands of albumin attached to the egg shell
3. **Egg Shell**
   • Shell gland secrete CaCO3 which deposit around the egg • It is also impregnated with proteins • Thousands of tiny pores (40-50µm)are for exchange of gases • Inside shell two membranes • Air pocke
4. 

Q12: Explain Coelom Formation

• Protostomes: Coelom is formed by splitting of mesoderm. This pattern is called schizocoelous.

• **Deuterostome:** Coelom is formed by out pouching of primitive gut (archenteron). This pattern is called enterocoelous.

Q13: Consumer

• Organisms which can not synthesize their food from inorganic compounds • They eat other organisms to get nutrition • All animals are consumers

Q14: Primary Consumers

• They are also called herbivores • These are the animals which feed on plants • E.g., • insects • grain eating birds • reptiles, birds and mammals which feed on leaves and other vegetation

Q15: Secondary Consumers

• Also called carnivores • These are the animals which feed on herbivores • E.g., lion, tiger, frog

Q16: Tertiary consumers

• These are the animals which feed on secondary consumers • E.g., a snake when eating a frog, a jaguar when hunting for a crocodile

Q17: Difference between Density independent factors and Density dependent factors

Population Density

• It is the number of individuals living per unit area. • Population density impact populations in predictable ways

Density-independent Factors

• These factors affect population regardless of their number • Weather conditions • Earth quake • Pollutants • Deforestation • Hurricane • Fire

Density-dependent Factors

These are the factors which are more severe when population density is high • Competition for resources • Disease • Predation • Parasitism

Q18:Spermatogenesis

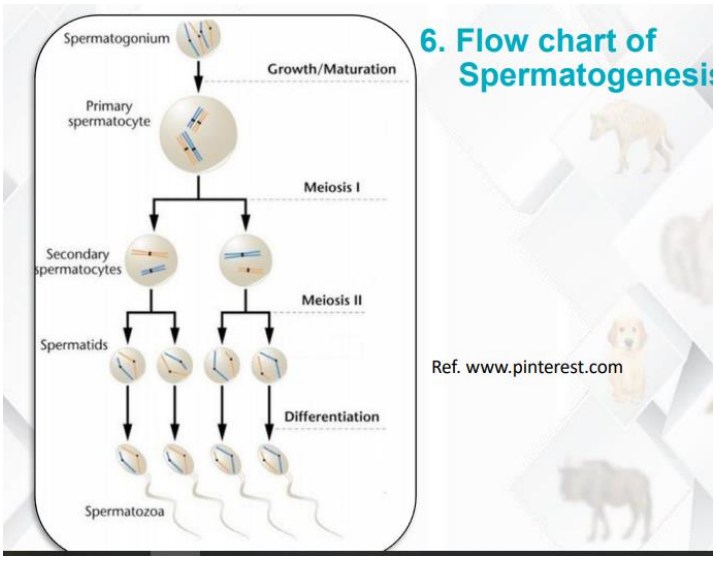
• The process of sperms production in male gonads through meiosis is called spermatogenesis

2. Location and duration • Spermatogenesis takes place in testes • In human the entire process of spermatogenesis and its transportation takes about 3 months • However, it is different in different animals

3. Spermatogoium and Primary Spermatocytes • Testes contain diploid stem cells • Sperm mother cell is called spermatogomium • It enlarges in its size and become primary spermatocyte

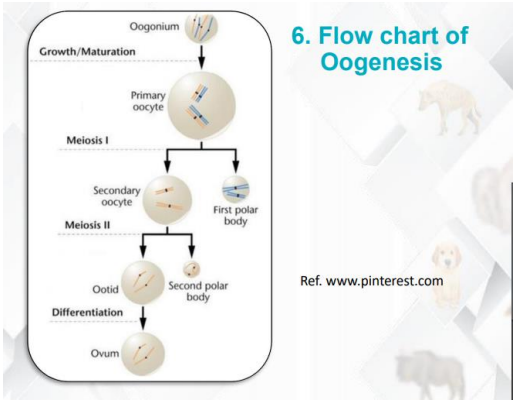
4. Secondary Spermatocytes and Spermatids • Primary spermatocyte undergo meiotic cell division. • After meiosis-I two secondary spermatocytes are formed • Meiosis-II results in four haploid spermatids

5. From spermatids to sperms • Spermatids do not undergo any further cell division • They undergo morphological changes to produce sperms



Q19: Gametogenesis OR Oogenesis

1. **Oogenesis**
   • The process of egg production in female gonads through meiosis is called oogenesis
2. **Location of Oogenesis**
   • Oogenesis takes place in ovaries • Follicles are specific structures in which egg formation and maturation takes place
3. **Oogoium and Primary Oocytes**
   • Ovaries contain oogonia. • Oogonium increase its size ….primary oocyte • Primary oocyte is diploid cell with duplicated chromosome • Each primary oocyte will form 1 ovum and polar bodies after reductive division
4. **Secondary Oocytes and Ovum**
   • Primary oocyte undergo meiotic cell division. • After meiosis-I one secondary oocyte and one polar body is formed • Meiosis-II results in a haploid ootid and another polar body• Ootid undergo differentiation process to form ovum



Q20: Differentiate between the Sprouter AND Seeders:

Types of Plants

Two types of plants

Sprouters: dormant meristem, storage and intact root help to regenerate quickly

Seeders: long lived seeds, germination induced by fire • Plants include toyon, chamise, poison oak, scrub oak, Yucca, Sumac

Q21: factores effecting on reproductive potential(5marks long)

Number of offspring produced

• Survival to reproductive age

• Reproductive period duration

• Length of time taken to maturity

Q22: plasmodstomes(not confirm)

Q23: Niche Diversification

• Competition between species may be reduced if resources are divided • One part of resources by one species and second part by other species • This mechanism is called niche diversification

Q24: Epigenesis Theory

• This theory gained popularity in mid 18th century • This theory says that embryo contain material from which embryo is gradually build • Much like a carpenter builds a building from nails, lumber and bricks

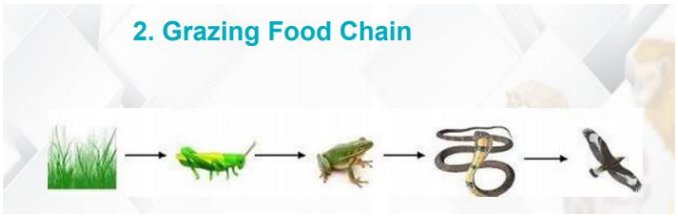
Q25: Food chain and its importance?

1. **Food Chain**

• It is the sequence of organisms in an ecosystem where energy moves from one organism to the next organism • It tells us the sequence to eat and being eaten

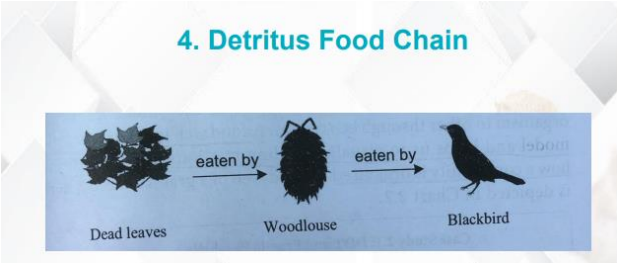
2. **Grazing Food Chain**

• It is the food chain that starts from the living green plants. • Green plants are eaten by herbivores which are eaten by carnivores • Such food chains are directly dependent on sunlight



3. **Detritus Food Chain**

• It is the food chain that starts from dead organic matter to microorganisms and then to organisms • They are little or not at all dependent on direct solar energy



4. **Importance of Food Chain**

• To understand the feeding relationships between organisms • To explore the energy flow mechanisms and matte circulation • To understand the problems of biomagnification

Q26:Neritic Ecosystem.2

• Neritic ecosystem extend from littoral zone to the edge of continental shelf • It is also called the shallow water

Q27: Enlist 5 types of ecological hierarchy?5

2. Ecological Hierarchy • It is the way in which ecological components interact with each other • There are five levels • Organism • Population • Community • Ecosystem • Biosphere

Organism

• Organism is the first level in ecological hierarchy • All the living things come at this level • E.g., human, lion, tiger, cow, tree, shrub • The way how organism interact with its environment is studied at this level

Population

• Group of organism of same species living at same geographical location at same time • At this level we study how the members of group live together and interact with each other

Community

• Groups of different species living together in a given area • The number of species represent the diversity of community • Community ecology include • Predation • Parasitism • Competition

Ecosystem

• It is the study of all population in a community and their physical environment • Biotic components and abiotic components • How living organism make the use of abiotic components • How they get energy • Effect of physical factors on organisms

Biosphere

• Biggest of all levels • All the area of earth and atmosphere occupied by living organisms • Main focus is on: • Geology • Climate • Water bodies • Latitude • Longitude

Q28: Disadvantage of sexual reproduction (5)

1. Advantages

a) Genetic Variability

• Genetic recombination • New combination of traits • This may help in changing environment or in case of disease

b) Foundation for Evolution

• Genetic variability is there • Variation is the foundation of evolution

c) Elimination of Deleterious Traits

• As there is genetic recombination • Deleterious mutation are eliminated

2. Disadvantages

a) Asexual animals

• Not feasible for animal which rely on asexual means only • Especially population with one sex only

b) Adaptive Gene Combination Dismantle

• In sexual reproduction new genetic combination arise • This will upset the already established adaptive gene combination

c) Wastage of energy

• Many of the gametes are not fertilized • This will cause a significant loss of energy

Q29: Difference in spiral and radial cleavage (3Mark's)

. **Spiral Cleavage** • It is pattern in which cells of animal pole are not exactly laying above the vegetal pole cells • Cleavage takes place at oblique angle forming a spiral pattern • Example: protostomes

Radial Cleavage • It is pattern in which cells of animal pole are exactly laying above the vegetal pole cells • Upper tier of cells is exactly on the lower tier of cells • Example: deurostomes

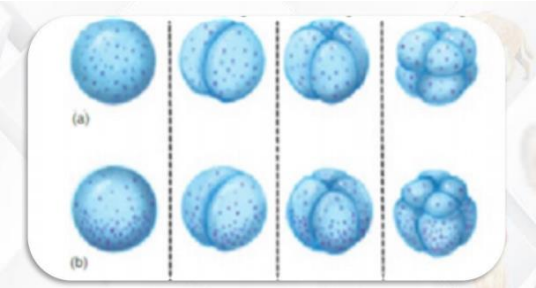
Q30: Explain meroblastic cleavage and Holoblastic Cleavage? 3marks

Meroblastic Cleavage

• Cleavage type in which egg is not completely divided into two cells • This is due to large amount of yolk • These embryo develop around or on top of yolk • Eggs of birds and reptiles

Holoblastic cleavage

• Cleavage type in which egg is completely divided into two cells • Egg contain little or moderate amount of yolk • Eggs of Amphibians, mammals, echinoderms, flatworms, nematodes



Q31: Difference between . Ectosymbiosis and Endosymbiosis?

Ectosymbiosis

It is a symbiosis in which symbionts live on the body surface of the host or in the digestive tact • E.g., Lice, barnacles and cleaner fish

Endosymbiosis

It is the relationship in which one symbiont live within the tissue of other. • E.g., nitrogen fixing bacteria in root nodules, algae inside the corals

Q32: Discuss coriolis effects? 3marks

Coriolis Effect

• Biomes are distinct by climate, pattern of rainfall, temperature and solar radiations • Warm air at equator rises and move towards poles and cold air away from pole • The earth rotation deflect this air to right and left of the poles • The effect is called Coriolis effect

Q33: Primary Succession

• The process of succession in an area that was not supporting any life previously is called primary succession. • Ecological succession in a lake formed by glacier or land exposed by land sliding

Q34: Birth of modern embryology (2)

• Modern embryology emerged with the proposed germ layer theory • Embryo form 2-3 layers from which different body parts are formed

Q35: Deffernece. b/w. phtoautotrophes and chemtrouphes

Photoautotrophs

They use sunlight as energy source for the synthesis of food from inorganic compounds • Green plants, algae, and photosynthetic bacteria • Photosynthesis transform less than 1% of sunlight reaching to earth • 30% reflects back • 20% is absorbed by atmosphere • 50% absorbed by earth

Chemolithotrophs

They obtain energy by oxidizing inorganic compounds • Usually live in extreme environments • E.g., sulfur bacteria

Q36:Ecological Pyramids

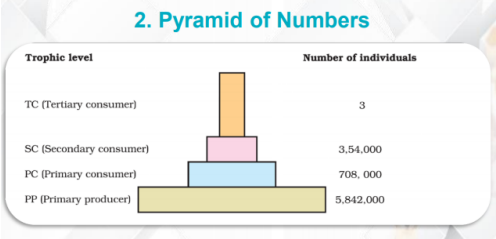
• It is the graphical representation of relationship between organisms at different trophic levels of an ecosystem • Three most common types of pyramids: • Pyramid of number • Pyramid of biomass • Pyramid of energy

Pyramid of Numbers

• It is the graphical representation of total number of different organisms present at each trophic level • This shows the population of organisms in a food chain

Pyramid of Numbers Example

• In a grass land the survey was conducted to check the trophic structure. • Following data was obtained • Producers: 1.5 x 106 • Primary consumers: 2x105 • Secondary consumers: 9x104 • Tertiary consumers: 1

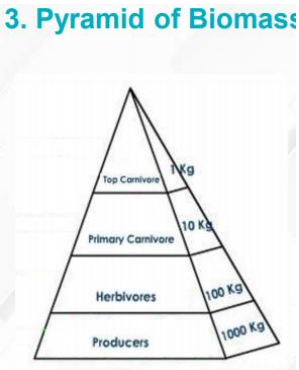


Pyramid of Biomass

• It is the representation of total mass of living matter present at each trophic level. • Biomass decrease at each ascending trophic level • Biomass is measured in grams of dry mass per square meter

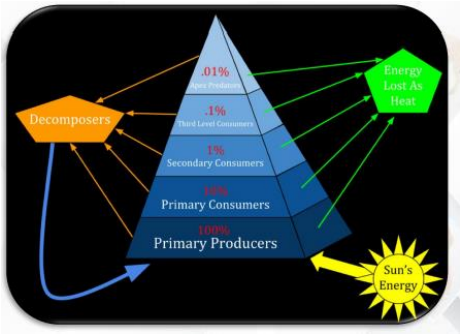
Pyramid of Biomass Example

• In a pond the survey was conducted to check the trophic structure. • Following data was obtained • Producers: 809 g/m2 • Primary consumers: 37 g/m2 • Secondary consumers: 11 g/m2 • Tertiary consumers: 1.5 g/m2



Pyramid of Energy

• It is the representation of total amount of energy present at each trophic level in an ecosystem • It is always upright • Energy pyramids help to compare the productivity of ecosystems.



Q37: Biodiversity loss (5)…

Biodiversity of Earth

• More than 1.6 million • Estimate is 4 to 6 million • More accurate is between 8.7±1.3million

Threats to Biodiversity

• Biodiversity of all natural areas is threatened • Acid rain • Pollution • Urbanization • Agriculture • Human is exploiting 40% of earth net primary production • Habitat loss

Solutions

• Realization about biodiversity • 22nd May is World Biodiversity Day • A system of international ethics that value natural diversity

Q38: Define Decomposers

• These are the organisms which obtain energy by breaking down dead organic matter into simpler compounds • Also called saprotrophs

Q39: Write notes on energy flow? 5marks

1. **Energy Budget**

• It is the balance sheet of energy gained and energy expenditure of an organism • An organism is an open thermodynamic system which exchange energy with its surrounding in three ways i.e., heat, work and potential energy of biomolecules • Energy budget allocation varies for ectotherms and endotherms. • Ectotherms rely on environment as heat source • Endotherms rely on metabolic heat to maintain body temperature

2. **Gross Productivity**

• Total amount of energy assimilated by an organism when an animal eats food • When an animal eats, food passes through its gut and nutrients are absorbed

3. **Net Productivity**

• Most of the energy assimilated is used in metabolic demands which include cellular metabolism and regulation of body temperature • So, net productivity is the amount of energy store in the form of biomass

4. Energy Flow

• Organisms require energy in one form or the other • Energy flow in an ecosystem is major factor for survival for great number of organisms • Source of energy is sunlight • Energy flows via food chain and food web • Flow is by 10% law • Energy flow is unidirectional

5. **Flow of material**

• All elements essential for life are derived from environment • They make all form of molecules for life. • When animals die and their dead bodies are decomposed, these materials are recycled in the ecosystem • Nutrient flow in ecosystem in a cyclic way